

1.9.4 Normal Forms for First Order Logic

Definition 5: A formula F in the first order logic is said to be in a prenex normal form if and only if the formula F is in the form of $(Q_1x_1) \dots (Q_nx_n)(M)$ where every (Q_ix_i) , $i=1, \dots, n$ is either $(\forall x_i)$ or $(\exists x_i)$, and M is a formula containing no quantifiers. $(Q_1x_1) \dots (Q_nx_n)$ is called the prefix and M is called the matrix of the formula F .

Example 7: $(\forall x)(\forall y)(P(x,y) \wedge Q(y)) \rightarrow R(z)$ are in prenex normal form.

- $\neg \forall x P(x) \rightarrow \exists x \neg P(x)$ (1)
 - $\neg \exists x P(x) \rightarrow \forall x \neg P(x)$ (2)
- } De Morgan's Laws for Quantifiers